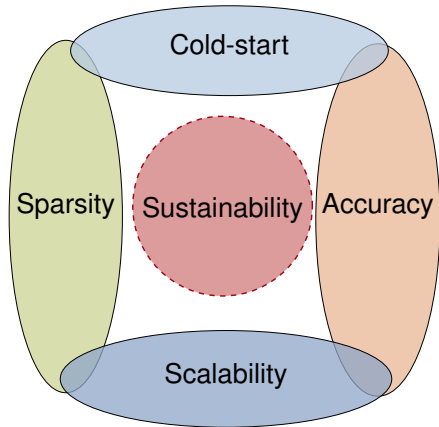


Energy-Aware Hybrid Recommender Systems Across the User Lifecycle

Saina Amiri Moghadam

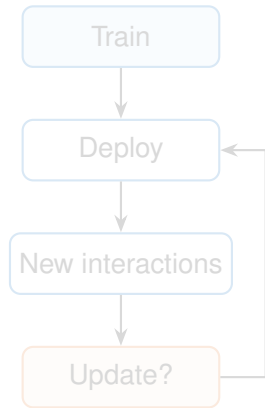
Chair of Connected Mobility
TUM School of Computation, Information and
Technology
Technical University of Munich

October 6, 2026



Introduction

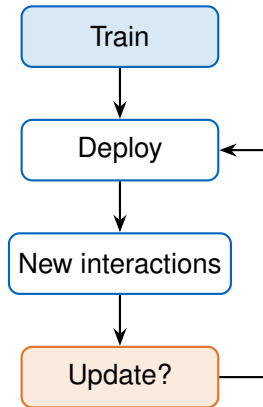
- Recommender models are becoming more expensive without proportionate gains in quality¹
- A recommender is not trained once: new users arrive and preferences shift, so the model must be kept up to date
- The recurring carbon cost of keeping it current has not been measured



¹G. Spillo et al., in Proceedings of the 17th acm conference on recommender systems, RecSys '23 (2023), pp. 856–862
T. Vente et al., in Proceedings of the 18th acm conference on recommender systems, RecSys '24 (2024), pp. 580–590

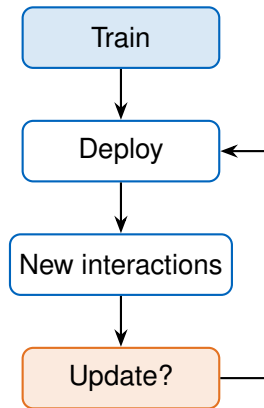
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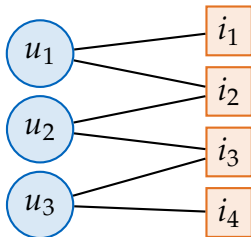
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What does it cost to keep a recommender up to date, and can a cheaper hybrid mechanism deliver comparable quality?



User–Item Interaction Graph

Neighborhood aggregation:

$$\mathbf{e}_u^{(k)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|} \sqrt{|\mathcal{N}_i|}} \mathbf{e}_i^{(k-1)}$$

Prediction: $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$

¹X. He et al., in Proceedings of the 43rd international acm sigir conference on research and development in information retrieval, SIGIR '20 (2020), pp. 639–648

CPU and RAM energy estimated
with CodeCarbon

$$\text{emissions} = C \cdot E$$

$$C = \sum_{s \in S} e_s \cdot p_s$$

$$C (\text{Germany}) = 381 \text{ g CO}_2\text{eq/kWh}$$

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Evaluation Metrics

Per batch: score all items,
mask items already seen,
recommend the top 10

Relevant: items the user
liked in this batch

$$\text{Precision@10} = \frac{\text{relevant items retrieved}}{10}$$

$$\text{Recall@10} = \frac{\text{relevant items retrieved}}{\text{relevant items}}$$

$$\text{NDCG@10} = \frac{\text{hits, weighted by rank}}{\text{best possible ranking}}$$

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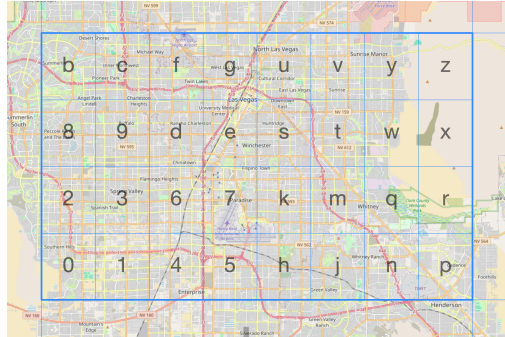
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Content Signal: Geohash



G. Niemeyer, *Geohash.org is public!*, (2008)

<https://web.archive.org/web/20080305223755/http://blog.labix.org/#post-84>

Incremental Update:

Keeps trained users up to
date

Content-Based Initialization:

Places users the model has
never seen



Combine both and compare quality against carbon
emissions

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Methodology

- Split Yelp and MovieLens along the global timeline: 80% training, 20% streamed in batches
- Compare no-update, incremental, content-based initialization, their combination and full retraining
- Measure Recall@10 per batch and emissions per phase with CodeCarbon

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Content-Based Initialization

- Profile trained users by location (geohash) and category counts
- Score a new user against them: $S = 0.7 \cdot \text{geo} + 0.3 \cdot \text{category}$
- Initialize with the weighted average of the top-20 most similar users' embeddings

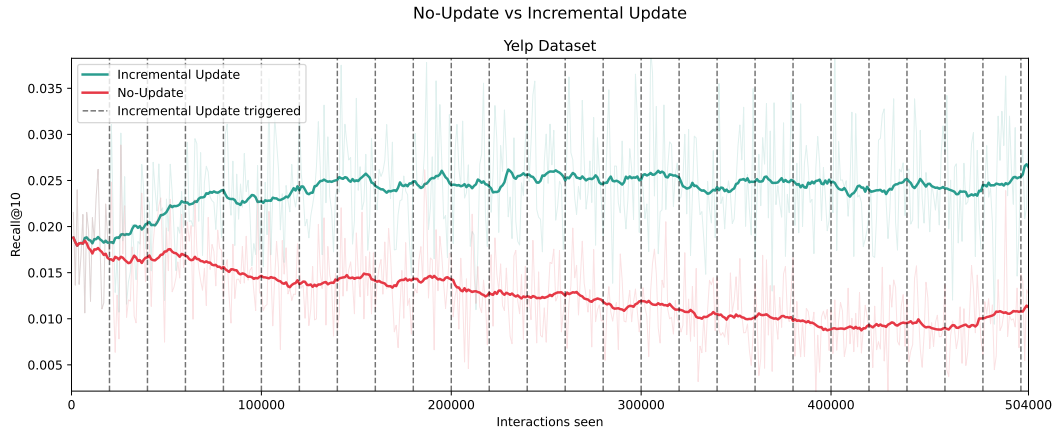
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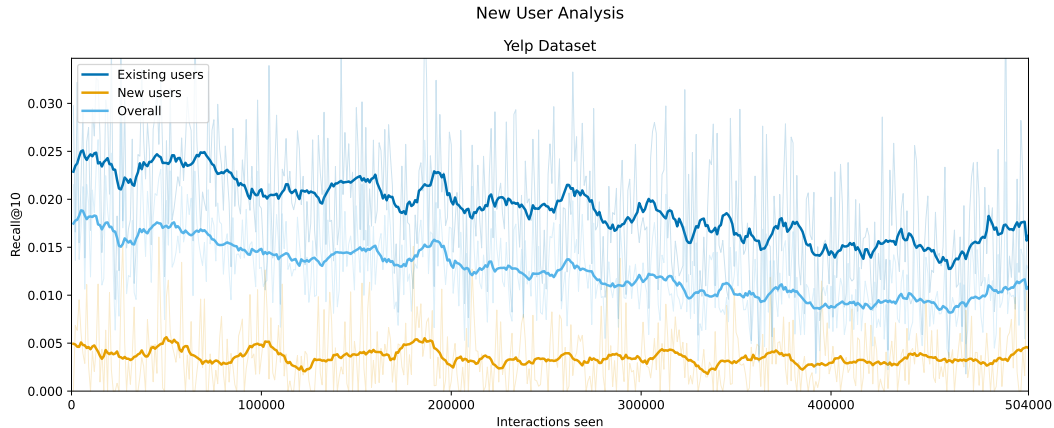
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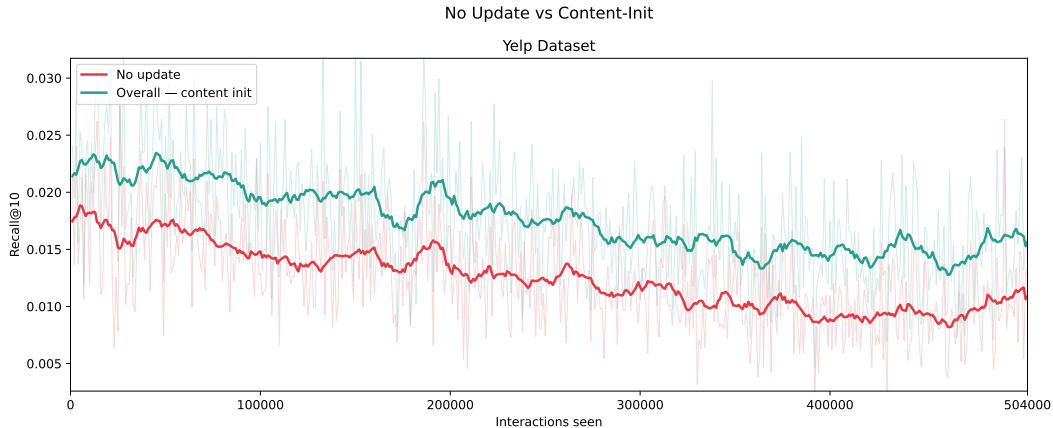
Yelp: No-Update vs. Incremental Update



Yelp: Existing vs. New Users

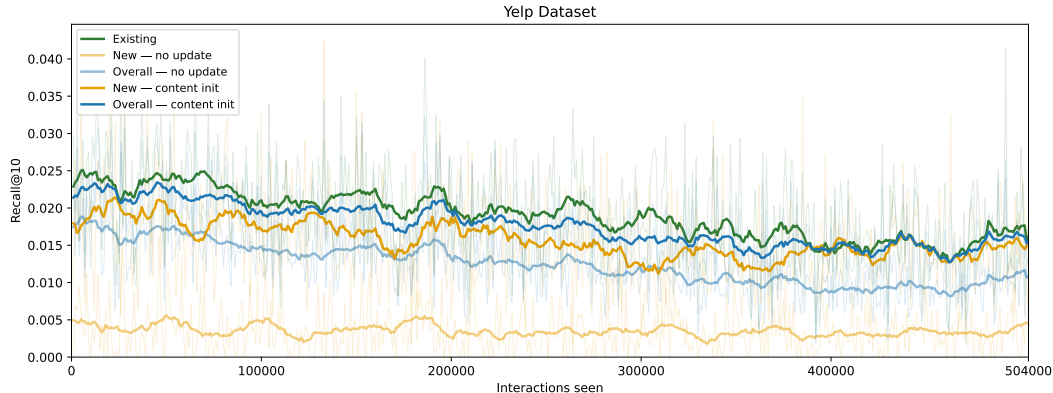


Yelp: Content-Based Initialization



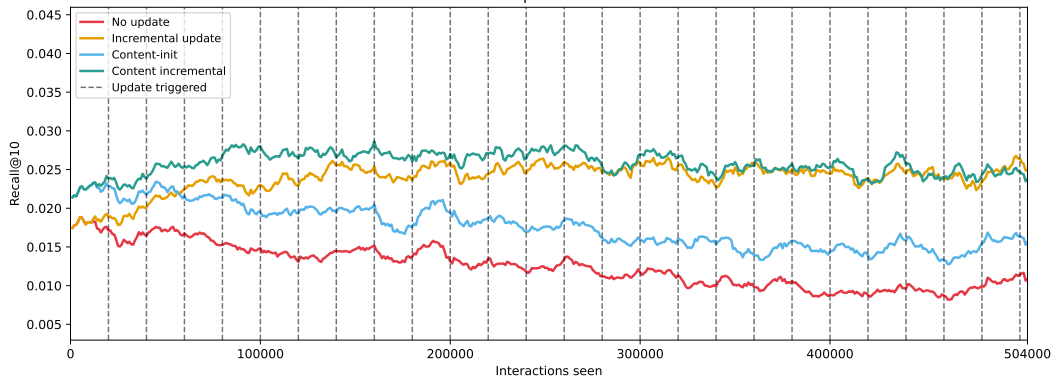
Yelp: Content-Based Initialization

No Update vs Content-Init by User Group

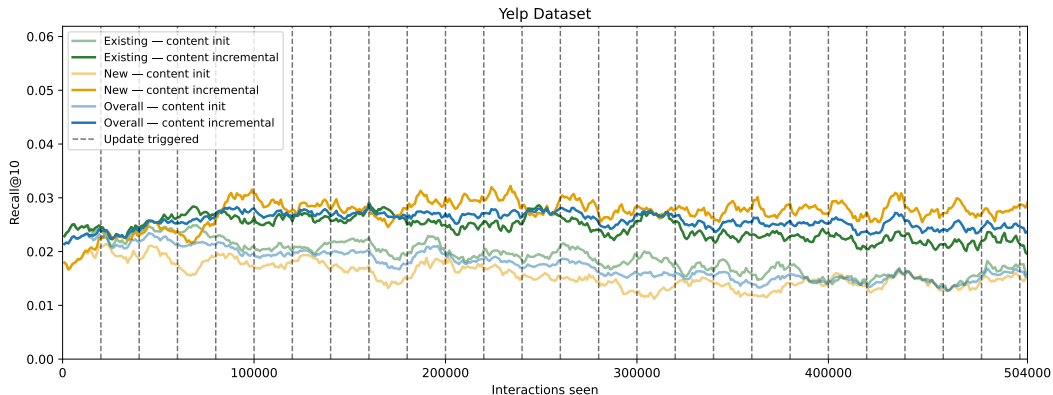


Strategy Comparison

Yelp Dataset



Content-Init vs Content Incremental by User Group



	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

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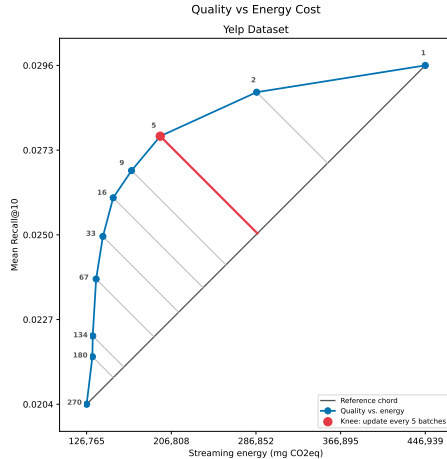
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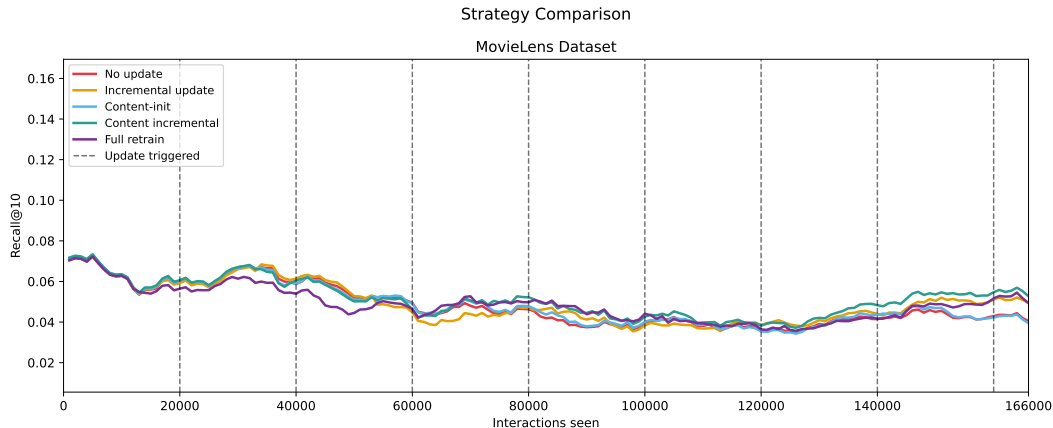
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Yelp: Update Frequency vs. Emissions



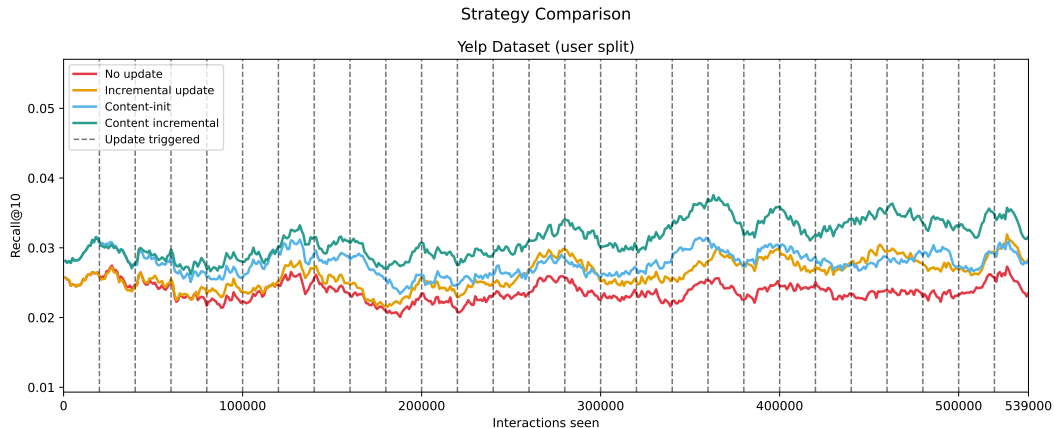
¹V. Satopää et al., in 2011 31st international conference on distributed computing systems workshops (2011), pp. 166–171

MovieLens: All Strategies



¹F. M. Harper and J. A. Konstan, ACM Transactions on Interactive Intelligent Systems 5, 19 (2015)

Pitfall: Per-User Split



¹Y. Ji et al., ACM Transactions on Information Systems **41**, 75, 1 (2023)

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¹Y. Ji et al., ACM Transactions on Information Systems **41**, 75, 1 (2023)

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Diagnose the source of decay before choosing an update strategy

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Next Steps

- Causal incremental graph convolution (CI-LightGCN), measured in energy rather than time²
- Windowed graph construction to bound the update cost³
- Streams with controlled, documented drift⁴
- More datasets with less strict interaction filtering

²S. Ding et al., IEEE Transactions on Neural Networks and Learning Systems **35**, 4718 (2024)

³A. Ghiye et al., in Proceedings of the 5th acm international conference on ai in finance, ICAIF '24 (2024), pp. 231–238

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Thank you!

Chair of Connected Mobility

Examiner: Prof. Dr.-Ing. Jörg Ott

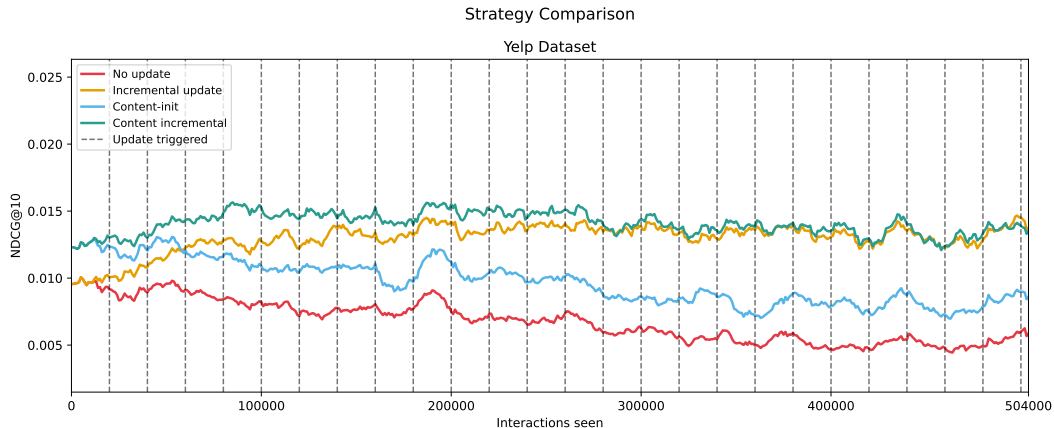
Advisor: Prof. Wolfgang Wörndl

Appendix

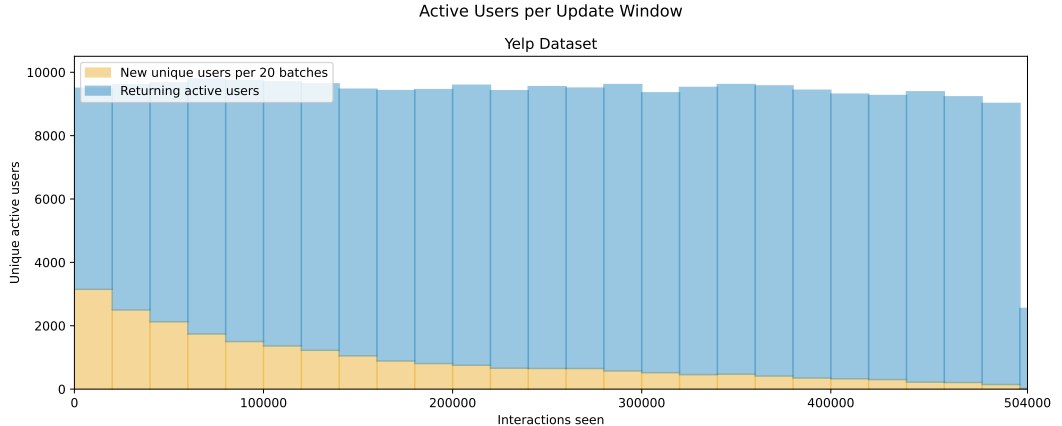
Yelp: Precision@10, All Strategies



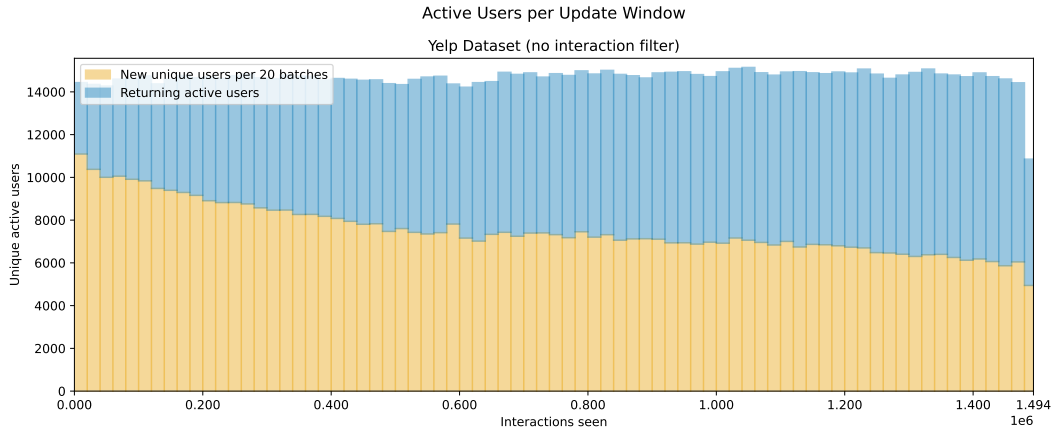
Yelp: NDCG@10, All Strategies



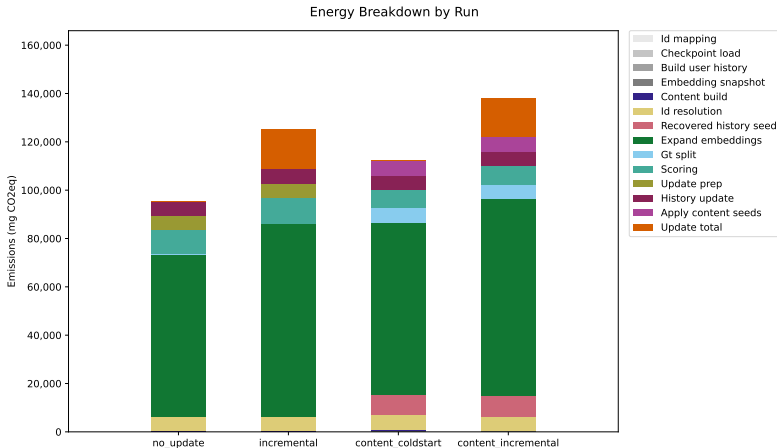
Yelp: New Unique Users per Update Window



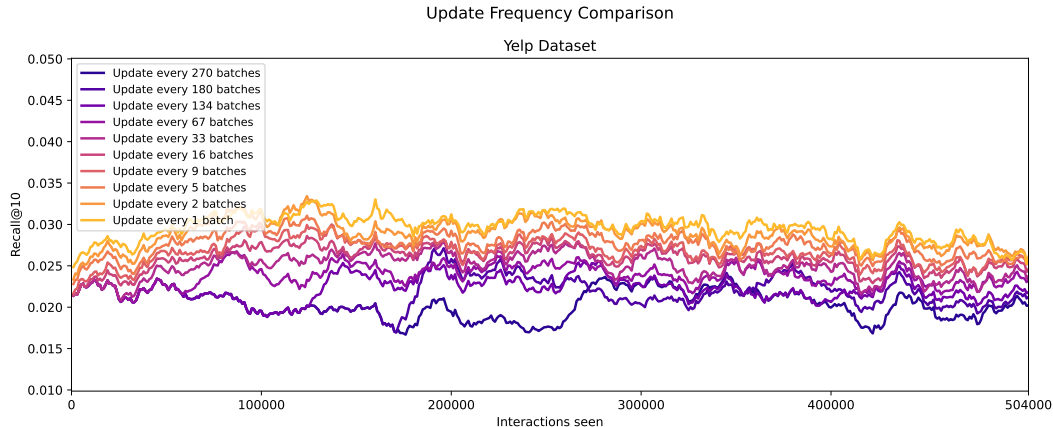
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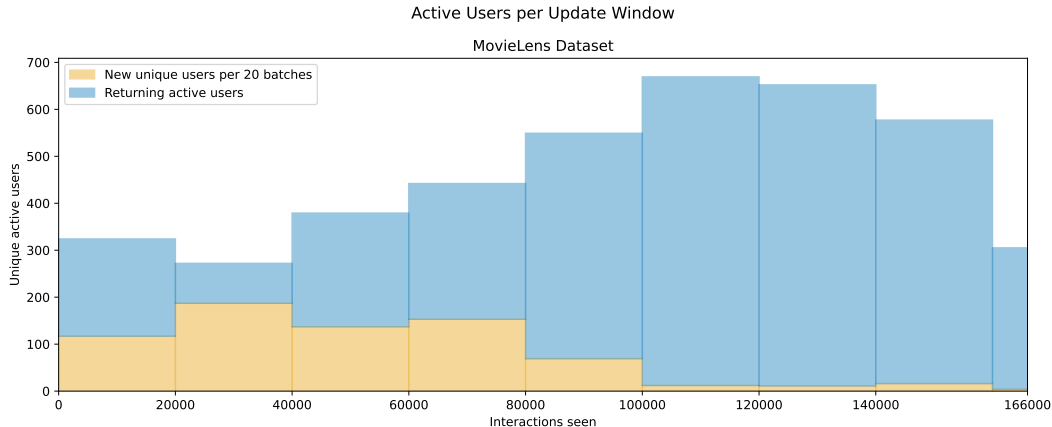
Yelp: Emissions by Phase



Yelp: Update Frequency Variants



MovieLens: New Unique Users per Update Window



Carbon Cost per Strategy

	No-Update	Incremental	Content-Init	Content + Incremental
Yelp streaming	95,306	125,053	112,422	138,321
Yelp training		1,832,436		
MovieLens streaming	17,286	18,654	20,045	22,618
MovieLens training		188,839		

Carbon emissions in mg CO₂eq. Full retrain on MovieLens: 1,387,568 streaming, 792× incremental per update.